

SoilLoop-2: Part Beta

Sensitivity analysis using definitions from Part Alpha

1. Sensitivity design

The sensitivity coefficient measures local change in final soil organic carbon after one parameter is perturbed. It is computed separately for S0, S1, and S2 and does not combine scenario probabilities.

$$S_x = (x / C_T)(\partial C_T / \partial x)$$

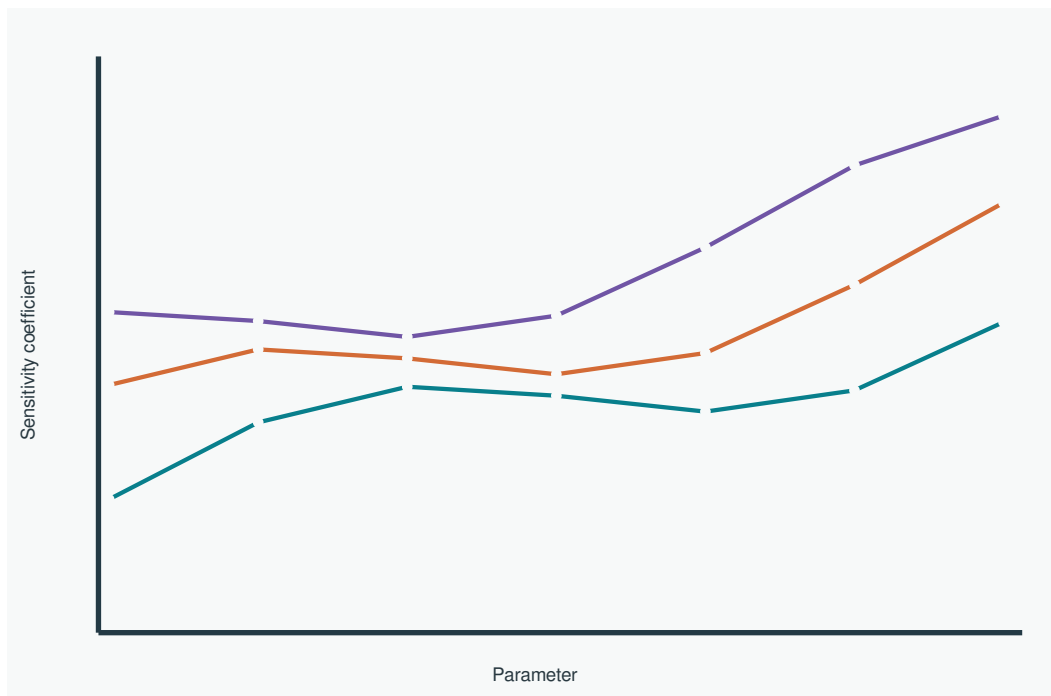


Figure 1. Local sensitivity coefficients for the three moisture response scenarios.

2. Interpretation

Input I dominates the final soil organic carbon value in all scenarios. The sensitivity coefficient for k_p grows under S2 because carbon remains longer in the protected pool.

Appendix note

Turnover time and moisture response retain the exact definitions supplied in Part Alpha. The accession SLC-26-09 identifies both parts as one closed synthetic corpus.

Footnote B. Derivatives use a symmetric 1 % perturbation around each listed parameter.

References

- [1] Kim H. Local sensitivity in coupled carbon pools. *Environmental Models* 20:120-131.
- [2] Park J. Conditional turnover time in two-pool models. *Soil Systems* 14:61-70.